

# Week 2 Linear Algebra

By

Mr. Farhan Ali

[farhan.ali@bnu.edu.pk](mailto:farhan.ali@bnu.edu.pk)

Beaconhouse National University, Lahore



## 1.4

# Inverses; Algebraic Properties of Matrices

In this section we will discuss some of the algebraic properties of matrix operations. We will see that many of the basic rules of arithmetic for real numbers hold for matrices, but we will also see that some do not.

# Properties of Matrix Addition and Scalar Multiplication

The following theorem lists the basic algebraic properties of the matrix operations.

## Theorem 1.4.1

### Properties of Matrix Arithmetic

Assuming that the sizes of the matrices are such that the indicated operations can be performed, the following rules of matrix arithmetic are valid.

- (a)  $A + B = B + A$  [Commutative law for matrix addition]
- (b)  $A + (B + C) = (A + B) + C$  [Associative law for matrix addition]
- (c)  $A(BC) = (AB)C$  [Associative law for matrix multiplication]
- (d)  $A(B + C) = AB + AC$  [Left distributive law]
- (e)  $(B + C)A = BA + CA$  [Right distributive law]
- (f)  $A(B - C) = AB - AC$
- (g)  $(B - C)A = BA - CA$
- (h)  $a(B + C) = aB + aC$
- (i)  $a(B - C) = aB - aC$
- (j)  $(a + b)C = aC + bC$
- (k)  $(a - b)C = aC - bC$
- (l)  $a(bC) = (ab)C$
- (m)  $a(BC) = (aB)C = B(aC)$

### EXAMPLE 1 | Associativity of Matrix Multiplication

As an illustration of the associative law for matrix multiplication, consider

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \\ 0 & 1 \end{bmatrix}, \quad B = \begin{bmatrix} 4 & 3 \\ 2 & 1 \end{bmatrix}, \quad C = \begin{bmatrix} 1 & 0 \\ 2 & 3 \end{bmatrix}$$

Then

$$AB = \begin{bmatrix} 1 & 2 \\ 3 & 4 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 4 & 3 \\ 2 & 1 \end{bmatrix} = \begin{bmatrix} 8 & 5 \\ 20 & 13 \\ 2 & 1 \end{bmatrix} \quad \text{and} \quad BC = \begin{bmatrix} 4 & 3 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 2 & 3 \end{bmatrix} = \begin{bmatrix} 10 & 9 \\ 4 & 3 \end{bmatrix}$$

Thus

$$(AB)C = \begin{bmatrix} 8 & 5 \\ 20 & 13 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 2 & 3 \end{bmatrix} = \begin{bmatrix} 18 & 15 \\ 46 & 39 \\ 4 & 3 \end{bmatrix}$$

and

$$A(BC) = \begin{bmatrix} 1 & 2 \\ 3 & 4 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 10 & 9 \\ 4 & 3 \end{bmatrix} = \begin{bmatrix} 18 & 15 \\ 46 & 39 \\ 4 & 3 \end{bmatrix}$$

so  $(AB)C = A(BC)$ , as guaranteed by Theorem 1.4.1(c).

## EXAMPLE 2 | Order Matters in Matrix Multiplication

Consider the matrices

$$A = \begin{bmatrix} -1 & 0 \\ 2 & 3 \end{bmatrix} \quad \text{and} \quad B = \begin{bmatrix} 1 & 2 \\ 3 & 0 \end{bmatrix}$$

Multiplying gives

$$AB = \begin{bmatrix} -1 & -2 \\ 11 & 4 \end{bmatrix} \quad \text{and} \quad BA = \begin{bmatrix} 3 & 6 \\ -3 & 0 \end{bmatrix}$$

Thus,  $AB \neq BA$ .

Because, as this example shows, it is *not* generally true that  $AB = BA$ , we say that ***matrix multiplication is not commutative***. This does not preclude the possibility of equality in certain cases—it is just not true in general. In those special cases where there is equality we say that  $A$  and  $B$  ***commute***.

## Zero Matrices

A matrix whose entries are all zero is called a **zero matrix**. Some examples are

$$\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}, \quad \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \quad \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}, \quad \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \quad [0]$$

We will denote a zero matrix by  $0$  unless it is important to specify its size, in which case we will denote the  $m \times n$  zero matrix by  $0_{m \times n}$ .

It should be evident that if  $A$  and  $0$  are matrices with the same size, then

$$A + 0 = 0 + A = A$$

Thus,  $0$  plays the same role in this matrix equation that the number  $0$  plays in the numerical equation  $a + 0 = 0 + a = a$ .

## Theorem 1.4.2

### Properties of Zero Matrices

If  $c$  is a scalar, and if the sizes of the matrices are such that the operations can be performed, then:

$$(a) \quad A + 0 = 0 + A = A$$

$$(b) \quad A - 0 = A$$

$$(c) \quad A - A = A + (-A) = 0$$

$$(d) \quad 0A = 0$$

$$(e) \quad \text{If } cA = 0, \text{ then } c = 0 \text{ or } A = 0.$$

### EXAMPLE 3 | Failure of the Cancellation Law

Consider the matrices

$$A = \begin{bmatrix} 0 & 1 \\ 0 & 2 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 1 \\ 3 & 4 \end{bmatrix}, \quad C = \begin{bmatrix} 2 & 5 \\ 3 & 4 \end{bmatrix}$$

We leave it for you to confirm that

$$AB = AC = \begin{bmatrix} 3 & 4 \\ 6 & 8 \end{bmatrix}$$

Although  $A \neq 0$ , canceling  $A$  from both sides of the equation  $AB = AC$  would lead to the incorrect conclusion that  $B = C$ . Thus, the cancellation law does not hold, in general, for matrix multiplication (though there may be particular cases where it is true).

## EXAMPLE 4 | A Zero Product with Nonzero Factors

Here are two matrices for which  $AB = 0$ , but  $A \neq 0$  and  $B \neq 0$ :

$$A = \begin{bmatrix} 0 & 1 \\ 0 & 2 \end{bmatrix}, \quad B = \begin{bmatrix} 3 & 7 \\ 0 & 0 \end{bmatrix}$$

## Identity Matrices

A square matrix with 1's on the main diagonal and zeros elsewhere is called an ***identity matrix***. Some examples are

$$[1], \quad \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \quad \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

An identity matrix is denoted by the letter  $I$ . If it is important to emphasize the size, we will write  $I_n$  for the  $n \times n$  identity matrix.

To explain the role of identity matrices in matrix arithmetic, let us consider the effect of multiplying a general  $2 \times 3$  matrix  $A$  on each side by an identity matrix. Multiplying on the right by the  $3 \times 3$  identity matrix yields

$$AI_3 = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \end{bmatrix} = A$$

and multiplying on the left by the  $2 \times 2$  identity matrix yields

$$I_2 A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \end{bmatrix} = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \end{bmatrix} = A$$

The same result holds in general; that is, if  $A$  is any  $m \times n$  matrix, then

$$AI_n = A \quad \text{and} \quad I_m A = A$$

Thus, the identity matrices play the same role in matrix arithmetic that the number 1 plays in the numerical equation  $a \cdot 1 = 1 \cdot a = a$ .

# Inverse of a Matrix

In real arithmetic every nonzero number  $a$  has a reciprocal  $a^{-1}(= 1/a)$  with the property

$$a \cdot a^{-1} = a^{-1} \cdot a = 1$$

The number  $a^{-1}$  is sometimes called the *multiplicative inverse* of  $a$ . Our next objective is to develop an analog of this result for matrix arithmetic. For this purpose we make the following definition.

## Definition 1

If  $A$  is a square matrix, and if there exists a matrix  $B$  of the same size for which  $AB = BA = I$ , then  $A$  is said to be ***invertible*** (or ***nonsingular***) and  $B$  is called an ***inverse*** of  $A$ . If no such matrix  $B$  exists, then  $A$  is said to be ***singular***.

The relationship  $AB = BA = I$  is not changed by interchanging  $A$  and  $B$ , so if  $A$  is invertible and  $B$  is an inverse of  $A$ , then it is also true that  $B$  is invertible, and  $A$  is an inverse of  $B$ . Thus, when  $AB = BA = I$  we say that  $A$  and  $B$  are *inverses of one another*.

## EXAMPLE 5 | An Invertible Matrix

Let

$$A = \begin{bmatrix} 2 & -5 \\ -1 & 3 \end{bmatrix} \quad \text{and} \quad B = \begin{bmatrix} 3 & 5 \\ 1 & 2 \end{bmatrix}$$

Then

$$AB = \begin{bmatrix} 2 & -5 \\ -1 & 3 \end{bmatrix} \begin{bmatrix} 3 & 5 \\ 1 & 2 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = I$$

$$BA = \begin{bmatrix} 3 & 5 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} 2 & -5 \\ -1 & 3 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = I$$

Thus,  $A$  and  $B$  are invertible and each is an inverse of the other.

## EXAMPLE 6 | A Class of Singular Matrices

A square matrix with a row or column of zeros is singular. To help understand why this is so, consider the matrix

$$A = \begin{bmatrix} 1 & 4 & 0 \\ 2 & 5 & 0 \\ 3 & 6 & 0 \end{bmatrix}$$

To prove that  $A$  is singular we must show that there is no  $3 \times 3$  matrix  $B$  such that

$$AB = BA = I$$

For this purpose let  $\mathbf{c}_1, \mathbf{c}_2, \mathbf{0}$  be the column vectors of  $A$ . Thus, for any  $3 \times 3$  matrix  $B$  we can express the product  $BA$  as

$$BA = B[\mathbf{c}_1 \quad \mathbf{c}_2 \quad \mathbf{0}] = [B\mathbf{c}_1 \quad B\mathbf{c}_2 \quad \mathbf{0}] \quad \text{[Formula (6) of Section 1.3]}$$

The column of zeros shows that  $BA \neq I$  and hence that  $A$  is singular.

The quantity  $ad - bc$  in Theorem 1.4.5 is called the **determinant** of the  $2 \times 2$  matrix  $A$  and is denoted by

$$\det(A) = ad - bc$$

or alternatively by

$$\begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc$$

$$\det(A) = \begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc$$

### Theorem 1.4.5

The matrix

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

is invertible if and only if  $ad - bc \neq 0$ , in which case the inverse is given by the formula

$$A^{-1} = \frac{1}{ad - bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix} \quad (2)$$

## Historical Note

The formula for  $A^{-1}$  given in Theorem 1.4.5 first appeared (in a more general form) in Arthur Cayley's 1858 *Memoir on the Theory of Matrices*. The more general result that Cayley discovered will be studied later.

## EXAMPLE 7 | Calculating the Inverse of a $2 \times 2$ Matrix

In each part, determine whether the matrix is invertible. If so, find its inverse.

$$(a) A = \begin{bmatrix} 6 & 1 \\ 5 & 2 \end{bmatrix} \quad (b) A = \begin{bmatrix} -1 & 2 \\ 3 & -6 \end{bmatrix}$$

**Solution (a)** The determinant of  $A$  is  $\det(A) = (6)(2) - (1)(5) = 7$ , which is nonzero. Thus,  $A$  is invertible, and its inverse is

$$A^{-1} = \frac{1}{7} \begin{bmatrix} 2 & -1 \\ -5 & 6 \end{bmatrix} = \begin{bmatrix} \frac{2}{7} & -\frac{1}{7} \\ -\frac{5}{7} & \frac{6}{7} \end{bmatrix}$$

We leave it for you to confirm that  $AA^{-1} = A^{-1}A = I$ .

**Solution (b)** The matrix is not invertible since  $\det(A) = (-1)(-6) - (2)(3) = 0$ .

## **EXAMPLE 8** | Solution of a Linear System by Matrix Inversion

A problem that arises in many applications is to solve a pair of equations of the form

$$u = ax + by$$

$$v = cx + dy$$

for  $x$  and  $y$  in terms of  $u$  and  $v$ . One approach is to treat this as a linear system of two equations in the unknowns  $x$  and  $y$  and use Gauss–Jordan elimination to solve for  $x$  and  $y$ . However, because the coefficients of the unknowns are *literal* rather than *numerical*, that procedure is a little clumsy. As an alternative approach, let us replace the two equations by the single matrix equation

$$\begin{bmatrix} u \\ v \end{bmatrix} = \begin{bmatrix} ax + by \\ cx + dy \end{bmatrix}$$

which we can rewrite as

$$\begin{bmatrix} u \\ v \end{bmatrix} = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$

If we assume that the  $2 \times 2$  matrix is invertible (i.e.,  $ad - bc \neq 0$ ), then we can multiply through on the left by the inverse and rewrite the equation as

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix}^{-1} \begin{bmatrix} u \\ v \end{bmatrix} = \begin{bmatrix} a & b \\ c & d \end{bmatrix}^{-1} \begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$

which simplifies to

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix}^{-1} \begin{bmatrix} u \\ v \end{bmatrix} = \begin{bmatrix} x \\ y \end{bmatrix}$$

Using Theorem 1.4.5, we can rewrite this equation as

$$\frac{1}{ad - bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix} \begin{bmatrix} u \\ v \end{bmatrix} = \begin{bmatrix} x \\ y \end{bmatrix}$$

from which we obtain

$$x = \frac{du - bv}{ad - bc}, \quad y = \frac{av - cu}{ad - bc}$$

### EXAMPLE 9 | The Inverse of a Product

Consider the matrices

$$A = \begin{bmatrix} 1 & 2 \\ 1 & 3 \end{bmatrix}, \quad B = \begin{bmatrix} 3 & 2 \\ 2 & 2 \end{bmatrix}$$

We leave it for you to show that

$$AB = \begin{bmatrix} 7 & 6 \\ 9 & 8 \end{bmatrix}, \quad (AB)^{-1} = \begin{bmatrix} 4 & -3 \\ -\frac{9}{2} & \frac{7}{2} \end{bmatrix}$$

and also that

$$A^{-1} = \begin{bmatrix} 3 & -2 \\ -1 & 1 \end{bmatrix}, \quad B^{-1} = \begin{bmatrix} 1 & -1 \\ -1 & \frac{3}{2} \end{bmatrix}, \quad B^{-1}A^{-1} = \begin{bmatrix} 1 & -1 \\ -1 & \frac{3}{2} \end{bmatrix} \begin{bmatrix} 3 & -2 \\ -1 & 1 \end{bmatrix} = \begin{bmatrix} 4 & -3 \\ -\frac{9}{2} & \frac{7}{2} \end{bmatrix}$$

Thus,  $(AB)^{-1} = B^{-1}A^{-1}$  as guaranteed by Theorem 1.4.6.

# Powers of a Matrix

If  $A$  is a *square* matrix, then we define the nonnegative integer powers of  $A$  to be

$$A^0 = I \quad \text{and} \quad A^n = AA \cdots A \quad [n \text{ factors}]$$

and if  $A$  is invertible, then we define the negative integer powers of  $A$  to be

$$A^{-n} = (A^{-1})^n = A^{-1}A^{-1} \cdots A^{-1} \quad [n \text{ factors}]$$

Because these definitions parallel those for real numbers, the usual laws of nonnegative exponents hold; for example,

$$A^r A^s = A^{r+s} \quad \text{and} \quad (A^r)^s = A^{rs}$$

In addition, we have the following properties of negative exponents.

## EXAMPLE 11 | The Square of a Matrix Sum

In real arithmetic, where we have a commutative law for multiplication, we can write

$$(a + b)^2 = a^2 + ab + ba + b^2 = a^2 + ab + ab + b^2 = a^2 + 2ab + b^2$$

However, in matrix arithmetic, where we have no commutative law for multiplication, the best we can do is to write

$$(A + B)^2 = A^2 + AB + BA + B^2$$

It is only in the special case where  $A$  and  $B$  *commute* (i.e.,  $AB = BA$ ) that we can go a step further and write

$$(A + B)^2 = A^2 + 2AB + B^2$$

# Matrix Polynomials

If  $A$  is a square matrix, say  $n \times n$ , and if

$$p(x) = a_0 + a_1x + a_2x^2 + \cdots + a_mx^m$$

is any polynomial, then we define the  $n \times n$  matrix  $p(A)$  to be

$$p(A) = a_0I + a_1A + a_2A^2 + \cdots + a_mA^m \quad (3)$$

where  $I$  is the  $n \times n$  identity matrix; that is,  $p(A)$  is obtained by substituting  $A$  for  $x$  and replacing the constant term  $a_0$  by the matrix  $a_0I$ . An expression of form (3) is called a ***matrix polynomial in  $A$*** .

## EXAMPLE 12 | A Matrix Polynomial

Find  $p(A)$  for

$$p(x) = x^2 - 2x - 5 \quad \text{and} \quad A = \begin{bmatrix} -1 & 2 \\ 1 & 3 \end{bmatrix}$$

**Solution**

$$\begin{aligned} p(A) &= A^2 - 2A - 5I \\ &= \begin{bmatrix} -1 & 2 \\ 1 & 3 \end{bmatrix}^2 - 2 \begin{bmatrix} -1 & 2 \\ 1 & 3 \end{bmatrix} - 5 \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \\ &= \begin{bmatrix} 3 & 4 \\ 2 & 11 \end{bmatrix} - \begin{bmatrix} -2 & 4 \\ 2 & 6 \end{bmatrix} - \begin{bmatrix} 5 & 0 \\ 0 & 5 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} \end{aligned}$$

or more briefly,  $p(A) = 0$ .

# Properties of the Transpose

The following theorem lists the main properties of the transpose.

## Theorem 1.4.8

If the sizes of the matrices are such that the stated operations can be performed, then:

$$(a) \quad (A^T)^T = A$$

$$(b) \quad (A + B)^T = A^T + B^T$$

$$(c) \quad (A - B)^T = A^T - B^T$$

$$(d) \quad (kA)^T = kA^T$$

$$(e) \quad (AB)^T = B^T A^T$$

### EXAMPLE 13 | Inverse of a Transpose

Consider a general  $2 \times 2$  invertible matrix and its transpose:

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \quad \text{and} \quad A^T = \begin{bmatrix} a & c \\ b & d \end{bmatrix}$$

Since  $A$  is invertible, its determinant  $ad - bc$  is nonzero. But the determinant of  $A^T$  is also  $ad - bc$  (verify), so  $A^T$  is also invertible. It follows from Theorem 1.4.5 that

$$(A^T)^{-1} = \begin{bmatrix} \frac{d}{ad - bc} & -\frac{c}{ad - bc} \\ -\frac{b}{ad - bc} & \frac{a}{ad - bc} \end{bmatrix}$$

which is the same matrix that results if  $A^{-1}$  is transposed (verify). Thus,

$$(A^T)^{-1} = (A^{-1})^T$$

as guaranteed by Theorem 1.4.9.

In Exercises **5–8**, use Theorem 1.4.5 to compute the inverse of the matrix.

5.  $A = \begin{bmatrix} 2 & -3 \\ 4 & 4 \end{bmatrix}$

6.  $B = \begin{bmatrix} 3 & 1 \\ 5 & 2 \end{bmatrix}$

7.  $C = \begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix}$

8.  $D = \begin{bmatrix} 6 & 4 \\ -2 & -1 \end{bmatrix}$

In Exercises **21–22**, compute  $p(A)$  for the given matrix  $A$  and the following polynomials.

a.  $p(x) = x - 2$

b.  $p(x) = 2x^2 - x + 1$

c.  $p(x) = x^3 - 2x + 1$

$$21. A = \begin{bmatrix} 3 & 1 \\ 2 & 1 \end{bmatrix}$$

$$22. A = \begin{bmatrix} 2 & 0 \\ 4 & 1 \end{bmatrix}$$

*In Exercises 25–28, use the method of Example 8 to find the unique solution of the given linear system.*

$$25. \begin{aligned} 3x_1 - 2x_2 &= -1 \\ 4x_1 + 5x_2 &= 3 \end{aligned}$$

$$26. \begin{aligned} -x_1 + 5x_2 &= 4 \\ -x_1 - 3x_2 &= 1 \end{aligned}$$

**1.5**

# Elementary Matrices and a Method for Finding $A^{-1}$

In this section we will develop an algorithm for finding the inverse of a matrix, and we will discuss some of the basic properties of invertible matrices.

## Elementary Matrices

In Section 1.1 we defined three elementary row operations on a matrix  $A$ :

1. Multiply a row by a nonzero constant  $c$ .
2. Interchange two rows.
3. Add a constant  $c$  times one row to another.

## Definition 2

A matrix  $E$  is called an **elementary matrix** if it can be obtained from an identity matrix by performing a *single* elementary row operation.

## EXAMPLE 4 | Using Row Operations to Find $A^{-1}$

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Find the inverse of

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 5 & 3 \\ 1 & 0 & 8 \end{bmatrix}$$

**Solution** We want to reduce  $A$  to the identity matrix by row operations and simultaneously apply these operations to  $I$  to produce  $A^{-1}$ . To accomplish this we will adjoin the identity matrix to the right side of  $A$ , thereby producing a partitioned matrix of the form

$$[A \mid I]$$

Then we will apply row operations to this matrix until the left side is reduced to  $I$ ; these operations will convert the right side to  $A^{-1}$ , so the final matrix will have the form

$$[I \mid A^{-1}]$$

The computations are as follows:

$$\left[ \begin{array}{ccc|ccc} 1 & 2 & 3 & 1 & 0 & 0 \\ 2 & 5 & 3 & 0 & 1 & 0 \\ 1 & 0 & 8 & 0 & 0 & 1 \end{array} \right]$$

$$\left[ \begin{array}{ccc|ccc} 1 & 2 & 3 & 1 & 0 & 0 \\ 0 & 1 & -3 & -2 & 1 & 0 \\ 0 & -2 & 5 & -1 & 0 & 1 \end{array} \right]$$

← We added  $-2$  times the first row to the second and  $-1$  times the first row to the third.

$$\left[ \begin{array}{ccc|ccc} 1 & 2 & 3 & 1 & 0 & 0 \\ 0 & 1 & -3 & -2 & 1 & 0 \\ 0 & 0 & -1 & -5 & 2 & 1 \end{array} \right]$$

← We added 2 times the second row to the third.

$$\left[ \begin{array}{ccc|ccc} 1 & 2 & 3 & 1 & 0 & 0 \\ 0 & 1 & -3 & -2 & 1 & 0 \\ 0 & 0 & 1 & 5 & -2 & -1 \end{array} \right]$$

← We multiplied the third row by  $-1$ .

The computations are as follows:

$$\left[ \begin{array}{ccc|ccc} 1 & 2 & 3 & 1 & 0 & 0 \\ 2 & 5 & 3 & 0 & 1 & 0 \\ 1 & 0 & 8 & 0 & 0 & 1 \end{array} \right]$$

$$\left[ \begin{array}{ccc|ccc} 1 & 2 & 3 & 1 & 0 & 0 \\ 0 & 1 & -3 & -2 & 1 & 0 \\ 0 & -2 & 5 & -1 & 0 & 1 \end{array} \right]$$

← We added  $-2$  times the first row to the second and  $-1$  times the first row to the third.

$$\left[ \begin{array}{ccc|ccc} 1 & 2 & 3 & 1 & 0 & 0 \\ 0 & 1 & -3 & -2 & 1 & 0 \\ 0 & 0 & -1 & -5 & 2 & 1 \end{array} \right]$$

← We added 2 times the second row to the third.

$$\left[ \begin{array}{ccc|ccc} 1 & 2 & 3 & 1 & 0 & 0 \\ 0 & 1 & -3 & -2 & 1 & 0 \\ 0 & 0 & 1 & 5 & -2 & -1 \end{array} \right]$$

← We multiplied the third row by  $-1$ .

$$\left[ \begin{array}{ccc|ccc} 1 & 2 & 0 & -14 & 6 & 3 \\ 0 & 1 & 0 & 13 & -5 & -3 \\ 0 & 0 & 1 & 5 & -2 & -1 \end{array} \right]$$

← We added 3 times the third row to the second and  $-3$  times the third row to the first.

$$\left[ \begin{array}{ccc|ccc} 1 & 0 & 0 & -40 & 16 & 9 \\ 0 & 1 & 0 & 13 & -5 & -3 \\ 0 & 0 & 1 & 5 & -2 & -1 \end{array} \right]$$

← We added  $-2$  times the second row to the first.

Thus,

$$A^{-1} = \begin{bmatrix} -40 & 16 & 9 \\ 13 & -5 & -3 \\ 5 & -2 & -1 \end{bmatrix}$$

## EXAMPLE 5 | Showing That a Matrix Is Not Invertible

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Consider the matrix

$$A = \begin{bmatrix} 1 & 6 & 4 \\ 2 & 4 & -1 \\ -1 & 2 & 5 \end{bmatrix}$$

Applying the procedure of Example 4 yields

$$\left[ \begin{array}{ccc|ccc} 1 & 6 & 4 & 1 & 0 & 0 \\ 2 & 4 & -1 & 0 & 1 & 0 \\ -1 & 2 & 5 & 0 & 0 & 1 \end{array} \right]$$

$$\left[ \begin{array}{ccc|ccc} 1 & 6 & 4 & 1 & 0 & 0 \\ 0 & -8 & -9 & -2 & 1 & 0 \\ 0 & 8 & 9 & 1 & 0 & 1 \end{array} \right]$$

← We added  $-2$  times the first row to the second and added the first row to the third.

$$\left[ \begin{array}{ccc|ccc} 1 & 6 & 4 & 1 & 0 & 0 \\ 0 & -8 & -9 & -2 & 1 & 0 \\ 0 & 0 & 0 & -1 & 1 & 1 \end{array} \right]$$

← We added the second row to the third.

Since we have obtained a row of zeros on the left side,  $A$  is not invertible.

## EXAMPLE 5 | Showing That a Matrix Is Not Invertible

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Consider the matrix

$$A = \begin{bmatrix} 1 & 6 & 4 \\ 2 & 4 & -1 \\ -1 & 2 & 5 \end{bmatrix}$$

Applying the procedure of Example 4 yields

$$\left[ \begin{array}{ccc|ccc} 1 & 6 & 4 & 1 & 0 & 0 \\ 2 & 4 & -1 & 0 & 1 & 0 \\ -1 & 2 & 5 & 0 & 0 & 1 \end{array} \right]$$

$$\left[ \begin{array}{ccc|ccc} 1 & 6 & 4 & 1 & 0 & 0 \\ 0 & -8 & -9 & -2 & 1 & 0 \\ 0 & 8 & 9 & 1 & 0 & 1 \end{array} \right]$$

← We added  $-2$  times the first row to the second and added the first row to the third.

$$\left[ \begin{array}{ccc|ccc} 1 & 6 & 4 & 1 & 0 & 0 \\ 0 & -8 & -9 & -2 & 1 & 0 \\ 0 & 0 & 0 & -1 & 1 & 1 \end{array} \right]$$

← We added the second row to the third.

Since we have obtained a row of zeros on the left side,  $A$  is not invertible.

## Exercise Set 1.5

In Exercises 1–2, determine whether the given matrix is elementary.

1. a.  $\begin{bmatrix} 1 & 0 \\ -5 & 1 \end{bmatrix}$

b.  $\begin{bmatrix} -5 & 1 \\ 1 & 0 \end{bmatrix}$

c.  $\begin{bmatrix} 1 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$

d.  $\begin{bmatrix} 2 & 0 & 0 & 2 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$

In Exercises 3–4, find a row operation and the corresponding elementary matrix that will restore the given elementary matrix to the identity matrix.

3. a.  $\begin{bmatrix} 1 & -3 \\ 0 & 1 \end{bmatrix}$

b.  $\begin{bmatrix} -7 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

c.  $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -5 & 0 & 1 \end{bmatrix}$

d.  $\begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$

In Exercises **11–12**, use the inversion algorithm to find the inverse of the matrix (if the inverse exists).

**11. a.** 
$$\begin{bmatrix} 1 & 2 & 3 \\ 2 & 5 & 3 \\ 1 & 0 & 8 \end{bmatrix}$$

**b.** 
$$\begin{bmatrix} -1 & 3 & -4 \\ 2 & 4 & 1 \\ -4 & 2 & -9 \end{bmatrix}$$

## Working with Technology

**T1.** It can be proved that if the partitioned matrix

$$\begin{bmatrix} A & B \\ C & D \end{bmatrix}$$

is invertible, then its inverse is

$$\begin{bmatrix} A^{-1} + A^{-1}B(D - CA^{-1}B)^{-1}CA^{-1} & -A^{-1}B(D - CA^{-1}B)^{-1} \\ -(D - CA^{-1}B)^{-1}CA^{-1} & (D - CA^{-1}B)^{-1} \end{bmatrix}$$

provided that all of the inverses on the right side exist. Use this result to find the inverse of the matrix

$$\begin{bmatrix} 1 & 2 & 1 & 0 \\ 0 & -1 & 0 & 1 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 3 & 3 \end{bmatrix}$$

## EXAMPLE 1 | Solution of a Linear System Using $A^{-1}$

Consider the system of linear equations

$$x_1 + 2x_2 + 3x_3 = 5$$

$$2x_1 + 5x_2 + 3x_3 = 3$$

$$x_1 + 8x_3 = 17$$

In matrix form this system can be written as  $A\mathbf{x} = \mathbf{b}$ , where

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 5 & 3 \\ 1 & 0 & 8 \end{bmatrix}, \quad \mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}, \quad \mathbf{b} = \begin{bmatrix} 5 \\ 3 \\ 17 \end{bmatrix}$$

In Example 4 of the preceding section, we showed that  $A$  is invertible and

$$A^{-1} = \begin{bmatrix} -40 & 16 & 9 \\ 13 & -5 & -3 \\ 5 & -2 & -1 \end{bmatrix}$$

By Theorem 1.6.2, the solution of the system is

$$\mathbf{x} = A^{-1}\mathbf{b} = \begin{bmatrix} -40 & 16 & 9 \\ 13 & -5 & -3 \\ 5 & -2 & -1 \end{bmatrix} \begin{bmatrix} 5 \\ 3 \\ 17 \end{bmatrix} = \begin{bmatrix} 1 \\ -1 \\ 2 \end{bmatrix}$$

or  $x_1 = 1, x_2 = -1, x_3 = 2$ .

## EXAMPLE 2 | Solving Two Linear Systems at Once

Solve the systems

$$(a) \quad x_1 + 2x_2 + 3x_3 = 4$$

$$2x_1 + 5x_2 + 3x_3 = 5$$

$$x_1 \quad \quad + 8x_3 = 9$$

$$(b) \quad x_1 + 2x_2 + 3x_3 = 1$$

$$2x_1 + 5x_2 + 3x_3 = 6$$

$$x_1 \quad \quad + 8x_3 = -6$$

**Solution** The two systems have the same coefficient matrix. If we augment this coefficient matrix with the columns of constants on the right sides of these systems, we obtain

$$\left[ \begin{array}{ccc|c|c} 1 & 2 & 3 & 4 & 1 \\ 2 & 5 & 3 & 5 & 6 \\ 1 & 0 & 8 & 9 & -6 \end{array} \right]$$

Reducing this matrix to reduced row echelon form yields (verify)

$$\left[ \begin{array}{ccc|c|c} 1 & 0 & 0 & 1 & 2 \\ 0 & 1 & 0 & 0 & 1 \\ 0 & 0 & 1 & 1 & -1 \end{array} \right]$$

It follows from the last two columns that the solution of system (a) is  $x_1 = 1, x_2 = 0, x_3 = 1$  and the solution of system (b) is  $x_1 = 2, x_2 = 1, x_3 = -1$ .

### EXAMPLE 3 | Determining Consistency by Elimination

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What conditions must  $b_1$ ,  $b_2$ , and  $b_3$  satisfy in order for the system of equations

$$\begin{aligned}x_1 + x_2 + 2x_3 &= b_1 \\x_1 \quad \quad + x_3 &= b_2 \\2x_1 + x_2 + 3x_3 &= b_3\end{aligned}$$

to be consistent?

**Solution** The augmented matrix is

$$\left[ \begin{array}{ccc|c} 1 & 1 & 2 & b_1 \\ 1 & 0 & 1 & b_2 \\ 2 & 1 & 3 & b_3 \end{array} \right]$$

which can be reduced to row echelon form as follows:

$$\left[ \begin{array}{ccc|c} 1 & 1 & 2 & b_1 \\ 0 & -1 & -1 & b_2 - b_1 \\ 0 & -1 & -1 & b_3 - 2b_1 \end{array} \right] \quad \leftarrow \begin{array}{l} -1 \text{ times the first row was added} \\ \text{to the second and } -2 \text{ times the} \\ \text{first row was added to the third.} \end{array}$$

$$\left[ \begin{array}{ccc|c} 1 & 1 & 2 & b_1 \\ 0 & 1 & 1 & b_1 - b_2 \\ 0 & -1 & -1 & b_3 - 2b_1 \end{array} \right] \quad \leftarrow \begin{array}{l} \text{The second row was} \\ \text{multiplied by } -1. \end{array}$$

$$\left[ \begin{array}{ccc|c} 1 & 1 & 2 & b_1 \\ 0 & 1 & 1 & b_1 - b_2 \\ 0 & 0 & 0 & b_3 - b_2 - b_1 \end{array} \right] \quad \leftarrow \begin{array}{l} \text{The second row was added} \\ \text{to the third.} \end{array}$$

It is now evident from the third row in the matrix that the system has a solution if and only if  $b_1$ ,  $b_2$ , and  $b_3$  satisfy the condition

$$b_3 - b_2 - b_1 = 0 \quad \text{or} \quad b_3 = b_1 + b_2$$

To express this condition another way,  $A\mathbf{x} = \mathbf{b}$  is consistent if and only if  $\mathbf{b}$  is a matrix of the form

$$\mathbf{b} = \begin{bmatrix} b_1 \\ b_2 \\ b_1 + b_2 \end{bmatrix}$$

where  $b_1$  and  $b_2$  are arbitrary.

## EXAMPLE 4 | Determining Consistency by Elimination

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What conditions must  $b_1$ ,  $b_2$ , and  $b_3$  satisfy in order for the system of equations

$$\begin{aligned}x_1 + 2x_2 + 3x_3 &= b_1 \\2x_1 + 5x_2 + 3x_3 &= b_2 \\x_1 \quad \quad + 8x_3 &= b_3\end{aligned}$$

to be consistent?

**Solution** The augmented matrix is

$$\left[ \begin{array}{ccc|c} 1 & 2 & 3 & b_1 \\ 2 & 5 & 3 & b_2 \\ 1 & 0 & 8 & b_3 \end{array} \right]$$

Reducing this to reduced row echelon form yields (verify)

$$\left[ \begin{array}{ccc|c} 1 & 0 & 0 & -40b_1 + 16b_2 + 9b_3 \\ 0 & 1 & 0 & 13b_1 - 5b_2 - 3b_3 \\ 0 & 0 & 1 & 5b_1 - 2b_2 - b_3 \end{array} \right] \quad (2)$$

In this case there are no restrictions on  $b_1$ ,  $b_2$ , and  $b_3$ , so the system has the unique solution

$$x_1 = -40b_1 + 16b_2 + 9b_3, \quad x_2 = 13b_1 - 5b_2 - 3b_3, \quad x_3 = 5b_1 - 2b_2 - b_3 \quad (3)$$

for all values of  $b_1$ ,  $b_2$ , and  $b_3$ .

## Exercise Set 1.6

*In Exercises 1–8, solve the system by inverting the coefficient matrix and using Theorem 1.6.2.*

1. 
$$\begin{aligned}x_1 + x_2 &= 2 \\ 5x_1 + 6x_2 &= 9\end{aligned}$$

2. 
$$\begin{aligned}4x_1 - 3x_2 &= -3 \\ 2x_1 - 5x_2 &= 9\end{aligned}$$

*In Exercises 9–12, solve the linear systems. Using the given values for the  $b$ 's solve the systems together by reducing an appropriate augmented matrix to reduced row echelon form.*

9. 
$$\begin{aligned}x_1 - 5x_2 &= b_1 \\ 3x_1 + 2x_2 &= b_2\end{aligned}$$

i.  $b_1 = 1, \quad b_2 = 4$

ii.  $b_1 = -2, \quad b_2 = 5$

In Exercises **13–17**, determine conditions on the  $b_i$ 's, if any, in order to guarantee that the linear system is consistent.

**13.** 
$$\begin{aligned}x_1 + 3x_2 &= b_1 \\ -2x_1 + x_2 &= b_2\end{aligned}$$

**14.** 
$$\begin{aligned}6x_1 - 4x_2 &= b_1 \\ 3x_1 - 2x_2 &= b_2\end{aligned}$$

**1.7**

## Diagonal, Triangular, and Symmetric Matrices

In this section we will discuss matrices that have various special forms. These matrices arise in a wide variety of applications and will play an important role in our subsequent work.

# Diagonal Matrices

A square matrix in which all the entries off the main diagonal are zero is called a ***diagonal matrix***. Here are some examples:

$$\begin{bmatrix} 2 & 0 \\ 0 & -5 \end{bmatrix}, \quad \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad \begin{bmatrix} 6 & 0 & 0 & 0 \\ 0 & -4 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 8 \end{bmatrix}, \quad \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

## EXAMPLE 1 | Inverses and Powers of Diagonal Matrices

If

$$A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & -3 & 0 \\ 0 & 0 & 2 \end{bmatrix}$$

then

$$A^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & -\frac{1}{3} & 0 \\ 0 & 0 & \frac{1}{2} \end{bmatrix}, \quad A^5 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & -243 & 0 \\ 0 & 0 & 32 \end{bmatrix}, \quad A^{-5} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & -\frac{1}{243} & 0 \\ 0 & 0 & \frac{1}{32} \end{bmatrix}$$

# Triangular Matrices

A square matrix in which all the entries above the main diagonal are zero is called **lower triangular**, and a square matrix in which all the entries below the main diagonal are zero is called **upper triangular**. A matrix that is either upper triangular or lower triangular is called **triangular**.

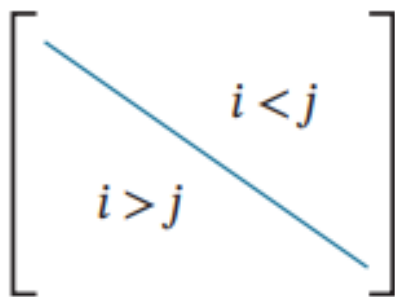


FIGURE 1.7.1

## EXAMPLE 2 | Upper and Lower Triangular Matrices

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} & a_{14} \\ 0 & a_{22} & a_{23} & a_{24} \\ 0 & 0 & a_{33} & a_{34} \\ 0 & 0 & 0 & a_{44} \end{bmatrix}$$

↑  
A general  $4 \times 4$  upper  
triangular matrix

$$\begin{bmatrix} a_{11} & 0 & 0 & 0 \\ a_{21} & a_{22} & 0 & 0 \\ a_{31} & a_{32} & a_{33} & 0 \\ a_{41} & a_{42} & a_{43} & a_{44} \end{bmatrix}$$

↑  
A general  $4 \times 4$  lower  
triangular matrix

# Properties of Triangular Matrices

Example 2 illustrates the following four facts about triangular matrices that we will state without formal proof:

- A square matrix  $A = [a_{ij}]$  is upper triangular if and only if all entries below the main diagonal are zero; that is,  $a_{ij} = 0$  if  $i > j$  (**Figure 1.7.1**).
- A square matrix  $A = [a_{ij}]$  is lower triangular if and only if all entries above the main diagonal are zero; that is,  $a_{ij} = 0$  if  $i < j$  (Figure 1.7.1).
- A square matrix  $A = [a_{ij}]$  is upper triangular if and only if the  $i$ th row starts with at least  $i - 1$  zeros for every  $i$ .
- A square matrix  $A = [a_{ij}]$  is lower triangular if and only if the  $j$ th column starts with at least  $j - 1$  zeros for every  $j$ .

# Symmetric Matrices

## Definition 1

A square matrix  $A$  is said to be *symmetric* if  $A = A^T$ .

## EXAMPLE 4 | Symmetric Matrices

The following matrices are symmetric since each is equal to its own transpose (verify).

$$\begin{bmatrix} 7 & -3 \\ -3 & 5 \end{bmatrix}, \quad \begin{bmatrix} 1 & 4 & 5 \\ 4 & -3 & 0 \\ 5 & 0 & 7 \end{bmatrix}, \quad \begin{bmatrix} d_1 & 0 & 0 & 0 \\ 0 & d_2 & 0 & 0 \\ 0 & 0 & d_3 & 0 \\ 0 & 0 & 0 & d_4 \end{bmatrix}$$

## Exercise Set 1.7

$$\begin{bmatrix} 1 & 4 & 5 \\ 4 & -3 & 0 \\ 5 & 0 & 7 \end{bmatrix}$$

In Exercises 1–2, classify the matrix as upper triangular, lower triangular, or diagonal, and decide by inspection whether the matrix is invertible. Recall that a diagonal matrix is both upper and lower triangular, so there may be more than one answer in some parts.

1. a.  $\begin{bmatrix} 2 & 1 \\ 0 & 3 \end{bmatrix}$

b.  $\begin{bmatrix} 0 & 0 \\ 4 & 0 \end{bmatrix}$

c.  $\begin{bmatrix} -1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & \frac{1}{5} \end{bmatrix}$

d.  $\begin{bmatrix} 3 & -2 & 7 \\ 0 & 0 & 3 \\ 0 & 0 & 8 \end{bmatrix}$

**Thank You**